

THE RIEMANN HYPOTHESIS AS A HERGLOTZ POSITIVITY: $\operatorname{Re} \xi'/\xi > 0$ IN THE RIGHT HALF-PLANE, AN UNCONDITIONAL BOUNDARY IDENTITY, AND THE HILBERT–PÓLYA REFORMULATION

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ABSTRACT. We give a clean, self-contained, computationally verified account of the de Branges chain criterion for the Riemann Hypothesis in its sharpest analytic form: RH is equivalent to the positivity $\operatorname{Re} \xi'/\xi(\sigma + i\tau) > 0$ for all $\sigma > \frac{1}{2}$, i.e. to ξ'/ξ being a Herglotz–Nevanlinna function on the right half-plane $\{\sigma > \frac{1}{2}\}$. We isolate one genuinely unconditional ingredient — the boundary identity $\operatorname{Re} \xi'/\xi(\frac{1}{2} + i\tau) = 0$ for all real τ , which follows from the functional equation alone — and use it to display the criterion as a harmonic-function problem: $\operatorname{Re} \xi'/\xi$ is harmonic in $\{\sigma > \frac{1}{2}\}$ away from the zeros, vanishes on the boundary, and grows at infinity, so the minimum principle yields positivity *provided there are no interior poles*; an off-line zero is exactly such a pole and forces a sign change. We record the resulting chain of equivalences $\text{RH} \iff \operatorname{Re} \xi'/\xi > 0 \iff \xi'/\xi \text{ Herglotz} \iff \text{Hilbert–Pólya}$ (a Herglotz function is the Weyl function of a self-adjoint operator, whose spectrum is the zeros, real iff RH). The criterion is classical in content (Hermite–Biehler / Laguerre–Pólya); our contribution is the clean unified statement, the explicit unconditional boundary identity, the harmonic-measure formulation of the obstruction, and full numerical verification. We are explicit that this reformulates RH and does not prove it: the missing step is the exclusion of interior poles from boundary data, the program’s wrong-sign capstone in harmonic-function form.

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1. INTRODUCTION

Let

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$$

be the completed Riemann zeta function: entire, of order 1, real on $\mathbb{R}$, with $\xi(s) = \xi(1-s)$, whose zeros are exactly the nontrivial zeros $\rho = \beta + i\gamma$ of ζ , all in the strip $0 < \beta < 1$. The Riemann Hypothesis is $\beta = \frac{1}{2}$ for all ρ .

A recurring theme of this line of work is that RH is a *positivity* of an indefinite (Kreĭn/Pontryagin) form, whose negative index counts the off-line zeros [1, 3, 4, 5]. The de Branges chain of Hilbert spaces of entire functions packages this positivity as the Hermite–Biehler property of a structure function. This paper extracts the resulting criterion in the cleanest possible analytic form and makes one of its ingredients fully unconditional and explicit.

The criterion. RH holds if and only if

$$(1) \quad \operatorname{Re} \frac{\xi'}{\xi}(\sigma + i\tau) > 0 \quad \text{for all } \sigma > \frac{1}{2}, \text{ all } \tau \in \mathbb{R},$$

i.e. ξ'/ξ is a Herglotz–Nevanlinna function on $\{\operatorname{Re} s > \frac{1}{2}\}$. The content of (1) is classical — it is equivalent to $\Xi(z) := \xi(\frac{1}{2} + iz)$ lying in the Laguerre–Pólya class (all zeros real), i.e. to the Hermite–Biehler property of the associated structure function [6, 7]. Our aims are (i) to state it cleanly and verify it; (ii) to isolate its one unconditional ingredient; (iii) to phrase the obstruction as a sharp harmonic-measure problem; and (iv) to record the Hilbert–Pólya equivalence transparently. We are explicit throughout that this is a *reformulation*, equivalent to RH, not a proof.

2. THE CRITERION AND THE OFF-LINE OBSTRUCTION

Proposition 2.1 (The criterion is RH). *Writing $\operatorname{Re} \xi'/\xi(\sigma + i\tau) = \sum_{\rho} \frac{\sigma - \beta_{\rho}}{|\sigma + i\tau - \rho|^2}$ (symmetric Hadamard sum over the zeros), the positivity (1) holds if and only if there is no zero with $\beta > \frac{1}{2}$ — i.e. if and only if RH.*

Proof. If RH holds, every term has $\sigma - \beta_{\rho} = \sigma - \frac{1}{2} > 0$ for $\sigma > \frac{1}{2}$, so $\operatorname{Re} \xi'/\xi > 0$. Conversely, suppose a zero $\rho_0 = \beta_0 + i\gamma_0$ has $\beta_0 > \frac{1}{2}$. At $s = \frac{1}{2} + \varepsilon + i\gamma_0$ with $0 < \varepsilon < \beta_0 - \frac{1}{2}$, its term is $\frac{\frac{1}{2} + \varepsilon - \beta_0}{(\frac{1}{2} + \varepsilon - \beta_0)^2} = \frac{-1}{\beta_0 - \frac{1}{2} - \varepsilon} < 0$; as $\varepsilon \downarrow 0$ this stays bounded away from 0, while the on-line zeros contribute $\frac{\varepsilon}{\varepsilon^2 + (\gamma_0 - \gamma)^2} \rightarrow 0$. Hence $\operatorname{Re} \xi'/\xi < 0$ at points arbitrarily close to $s = \frac{1}{2} + i\gamma_0$, violating (1). (Numerically, $\operatorname{Re} \xi'/\xi$ is small and positive between on-line zeros on $\sigma = \frac{1}{2} + \varepsilon$, so an $O(1)$ negative term flips the sign; Appendix A.) $\square$

Remark 2.2. Equation (1) says ξ'/ξ maps the half-plane $\{\operatorname{Re} s > \frac{1}{2}\}$ into the half-plane $\{\operatorname{Re} > 0\}$ — a *Herglotz–Nevanlinna* map. The zeros of ξ are the poles of ξ'/ξ (residue +1); a Herglotz function on a half-plane has all poles on the boundary, so (1) forces every zero onto $\sigma = \frac{1}{2}$.

3. THE UNCONDITIONAL BOUNDARY IDENTITY

The boundary of the half-plane carries an exact, RH-free fact.

Theorem 3.1 (Boundary identity). *For all real τ ,*

$$\operatorname{Re} \frac{\xi'}{\xi}\left(\frac{1}{2} + i\tau\right) = 0.$$

Proof. Differentiating $\xi(s) = \xi(1-s)$ gives $\xi'/\xi(s) = -\xi'/\xi(1-s)$. On $s = \frac{1}{2} + i\tau$ we have $1-s = \frac{1}{2} - i\tau = \bar{s}$, and since ξ has real Taylor coefficients, $\xi'/\xi(\bar{s}) = \overline{\xi'/\xi(s)}$. Therefore $\xi'/\xi(s) = -\overline{\xi'/\xi(s)}$, whence $\operatorname{Re} \xi'/\xi(s) = 0$. No use is made of the location of the zeros. (Verified to machine precision, Appendix A.) $\square$

Corollary 3.2 (Critical-line is a σ -critical line of $|\xi|$). $\partial_\sigma \log |\xi(\sigma + i\tau)| = \operatorname{Re} \xi'/\xi$ vanishes on $\sigma = \frac{1}{2}$; thus $\sigma \mapsto |\xi(\sigma + i\tau)|$ has a critical point at $\sigma = \frac{1}{2}$ for every τ . Its second derivative there is $\partial_\sigma \operatorname{Re} \xi'/\xi = \operatorname{Re} (\xi'/\xi)' = \sum_\rho \frac{(\tau - \gamma_\rho)^2 - (\frac{1}{2} - \beta_\rho)^2}{|\frac{1}{2} + i\tau - \rho|^4}$, which under RH equals $\sum_\rho (\tau - \gamma_\rho)^{-2} > 0$, so the critical line is a local minimum of $|\xi|$ in σ — the classical “ ξ is smallest on the critical line” picture.

4. THE HARMONIC-FUNCTION STRUCTURE AND THE OBSTRUCTION

$\operatorname{Re} \xi'/\xi$ is harmonic in $\{\sigma > \frac{1}{2}\}$ away from the zeros (as the real part of a meromorphic function), it vanishes on the boundary (Theorem 3.1), and $\operatorname{Re} \xi'/\xi(\sigma + i\tau) \rightarrow +\infty$ as $\sigma \rightarrow +\infty$ (the Γ -factor dominates).

Proposition 4.1 (Minimum principle vs. interior poles). *If ξ has no zero in $\{\sigma > \frac{1}{2}\}$ (i.e. RH), then $\operatorname{Re} \xi'/\xi$ is harmonic there, $= 0$ on the boundary and positive at infinity, so by the minimum principle $\operatorname{Re} \xi'/\xi > 0$ throughout $\{\sigma > \frac{1}{2}\}$. Conversely, a zero ρ_0 with $\beta_0 > \frac{1}{2}$ is an interior pole of ξ'/ξ with residue $+1$; near it $\operatorname{Re} \xi'/\xi \sim (\sigma - \beta_0)/|\sigma - \rho_0|^2$ takes both signs, so the positivity fails. Hence (1) $\iff$ no interior poles $\iff$ RH.*

Remark 4.2 (The new-tool target). Theorem 3.1 furnishes, unconditionally, the boundary data ($\operatorname{Re} = 0$ on $\sigma = \frac{1}{2}$) and the growth at infinity. What is missing is an argument that $\operatorname{Re} \xi'/\xi$ stays positive in $\frac{1}{2} < \sigma < 1$ without assuming the interior poles absent — a Phragmén–Lindelöf / harmonic-measure principle robust to possible interior singularities. Equivalently (next section), one must construct the self-adjoint operator whose Weyl function is ξ'/ξ . This is the open core.

5. THE HERGLOTZ = HILBERT–PÓLYA REFORMULATION

A function m holomorphic on a half-plane with $\operatorname{Re} m \geq 0$ there is a *Herglotz–Nevanlinna* function; by the spectral theorem such m are exactly the Weyl (m -) functions of self-adjoint operators / canonical systems, with the poles of m being the (real) eigenvalues [6, 7]. Combining:

$$\begin{aligned} \text{RH} &\iff \operatorname{Re} \frac{\xi'}{\xi} > 0 \text{ on } \{\sigma > \tfrac{1}{2}\} \iff \frac{\xi'}{\xi} \text{ is Herglotz} \\ &\iff \frac{\xi'}{\xi} \text{ is the Weyl function of a self-adjoint operator.} \end{aligned}$$

The last is Hilbert–Pólya: an operator $H = H^*$ with $\xi'/\xi(s) = \langle (\cdot - (s - \frac{1}{2})/i)^{-1} \rangle$ -type resolvent data, whose spectrum is $\{\gamma_\rho\}$, real iff RH. The chain makes explicit that the analytic positivity (1) and the spectral (Hilbert–Pólya) program are *the same statement*; the indefinite object carrying the off-line index is the de Branges/Pontryagin space $H(E_\xi)$ of the companion structure function [3, 5].

6. PLACE IN THE WRONG-SIGN CAPSTONE (HONEST STATUS)

The unconditional input is two-sided/boundary: $\operatorname{Re} \xi'/\xi = 0$ on the line (Theorem 3.1) and positive growth at infinity. The missing input is one-sided: the exclusion of interior negative singularities (Proposition 4.1). This is precisely the present work’s wrong-sign capstone [2]: every unconditional handle is a boundary/lower constraint, while RH is the one-sided/upper statement that no zero escapes the line. The de Branges sufficient condition for the chain to stay Hermite–Biehler was shown not to hold for ζ by Conrey–Li [8]; the finite-order truncations are each positive unconditionally, the full positivity is RH, and the gap is a uniformity gap [2]. We therefore present (1)–Theorem 3.1 as a clean, verified *reformulation* with one genuinely unconditional ingredient, not as progress past the wall.

CONCLUSION

RH is the Herglotz positivity of ξ'/ξ on the right half-plane; the critical line is, unconditionally, the locus where $\operatorname{Re} \xi'/\xi$ vanishes and where $|\xi|$ is σ -stationary; and the whole content is the exclusion of interior poles from boundary data — equivalently, the construction of a self-adjoint realization (Hilbert–Pólya). The reformulation is exact and clean; the crossing is the open problem.

APPENDIX A. REPRODUCIBLE COMPUTATIONS

A.1 The criterion $\operatorname{Re} \xi'/\xi > 0$ for $\sigma > \frac{1}{2}$ (Prop. 2.1).

```
import mpmath as mp
mp.mp.dps = 25
def xi(s): return mp.mpf('0.5')*s*(s-1)*mp.pi**(-s/2)*mp.gamma(s/2)*mp.zeta(s)
def xilog(s): return mp.diff(lambda z: mp.log(xi(z)), s) # xi'/xi
for sigma in ['1.0', '0.7', '0.55', '0.501']:
    sg=mp.mpf(sigma)
    vals=[mp.re(xilog(sg+1j*mp.mpf(t))) for t in [0,10,14.13,20,30,50]]
    print(sigma, "min Re xi'/xi =", mp.nstr(min(vals),5)) # all > 0 (RH holds in range)
```

A.2 The unconditional boundary identity $\operatorname{Re} \xi'/\xi(\frac{1}{2} + i\tau) = 0$ (Thm. 3.1).

```
import mpmath as mp
mp.mp.dps = 25
def xi(s): return mp.mpf('0.5')*s*(s-1)*mp.pi**(-s/2)*mp.gamma(s/2)*mp.zeta(s)
def xilog(s): return mp.diff(lambda z: mp.log(xi(z)), s)
for t in [1,5,10,14.13,20,50,100]:
    print(t, "Re xi'/xi(1/2+it) =", mp.nstr(mp.re(xilog(mp.mpf('0.5')+1j*mp.mpf(t))),4)) # = 0
```

A.3 The fine scan near the line (positivity is small between zeros).

```
import mpmath as mp
mp.mp.dps = 25
def xi(s): return mp.mpf('0.5')*s*(s-1)*mp.pi**(-s/2)*mp.gamma(s/2)*mp.zeta(s)
def xilog(s): return mp.diff(lambda z: mp.log(xi(z)), s)
sg=mp.mpf('0.51')
vals=[(float(t),float(mp.re(xilog(sg+1j*t)))) for t in mp.linspace(0.5,40,80)]
m=min(vals,key=lambda x:x[1]); print("min on sigma=0.51:", m) # small positive, between zeros
```

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